A bounded strictly starlike domain with a nonstarlike Cassinian ball

Principia Math in Progress

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Task 1. Resolve MathDB 331705

For a proper domain $G \subset \mathbb{R}^n$, its Cassinian metric is

$$c_G(x,y) = \sup_{p \in \partial G} \frac{|x-y|}{|x-p||p-y|}, \qquad B_{c_G}(x,r) = \{y \in G : c_G(x,y) < r\}.$$

Conjecture 4.7 of [1] asserts that if G is starlike with respect to x, then $B_{c_G}(x,r)$ is starlike with respect to x for every $r > 0$. The following counterexample also satisfies the stricter boundary-segment definition of starlikeness used in [2].

Theorem 1. *In $\mathbb{R}^2$, set*

$$K = \{(u,v) : u \geq 1,\ 4|v| \leq u - 1\}, \qquad G = B(0,10) \setminus K.$$

Then G is a bounded domain, strictly starlike with respect to 0, but $B_{c_G}(0,5/4)$ is not starlike with respect to 0.

Proof. The set K is closed, so G is open, bounded, and contains 0. For any $w = (u,v) \in \overline{G}$ and $0 \leq \tau < 1$, we have $|\tau w| < 10$. If $\tau u < 1$, then $\tau w \notin K$. Otherwise $u \geq 1$ and $w \notin \operatorname{int} K$ imply $4|v| \geq u - 1$, whence

$$4|\tau v| \geq \tau(u-1) = \tau u - \tau > \tau u - 1.$$

Again $\tau w \notin K$. Thus $[0,w) \subset G$ for all $w \in \overline{G}$; in particular G is path connected and strictly starlike about 0.

Take $y = (4,4)$ and $z = (1,1) = y/4$. Both points lie in G. We compute their exact Cassinian distances from 0. Every boundary point belongs either to the circle $|p| = 10$ or to one of the truncated rays

$$p_\pm(t) = (1+t,\ \pm t/4), \qquad t \geq 0.$$

For $w = (a,a)$ with $a \geq 0$, the squared distances satisfy

$$|p_-(t) - w|^2 - |p_+(t) - w|^2 = at \geq 0, \qquad |p_-(t)| = |p_+(t)|.$$

It suffices to bound the denominator on the upper ray. Put

$$P_a(t) = |p_+(t)|^2 |p_+(t) - (a,a)|^2 = \left((1+t)^2 + \frac{t^2}{16}\right)\left((1+t-a)^2 + \left(\frac{t}{4} - a\right)^2\right).$$

Direct expansion and completion of the square give

$$P_4(t) - 25 = 42t + t^2 \left[\frac{289}{256} \left(t - \frac{48}{17} \right)^2 + \frac{21}{8} \right] \geq 0,$$
$$P_1(t) - 1 = \frac{3}{2}t + \frac{9}{8}t^2 + \frac{51}{32}t^3 + \frac{289}{256}t^4 \geq 0.$$

Equality holds at $t = 0$, the actual boundary point $p = (1, 0)$. Therefore the supremum over the ray portions is $4\sqrt{2}/5$ for y, and $\sqrt{2}$ for z.

The circular portion cannot increase either value. Indeed, for $|p| = 10$,

$$\frac{|w|}{|p||p - w|} \leq \frac{|w|}{10(10 - |w|)}.$$

Since $|y| = 4\sqrt{2} < 6$ and $|z| = \sqrt{2} < 2$, this upper bound is respectively less than $3/20 < 4\sqrt{2}/5$ and $1/40 < \sqrt{2}$. Consequently

$$c_G(0, y) = \frac{4\sqrt{2}}{5} < \frac{5}{4}, \qquad c_G(0, z) = \sqrt{2} > \frac{5}{4}.$$

The squared comparisons are $32/25 < 25/16 < 2$. Thus $y \in B_{c_G}(0, 5/4)$ while the point $z = y/4$ of the segment $[0, y]$ is outside that ball, proving the assertion. □

Scope. This refutes the universal claim in dimension two, even with a bounded strictly starlike domain whose removed set has positive interior. It does not resolve the separate convex-domain conjecture adjacent to the selected statement in [1].

Task 2. Smooth the domain. Construct a counterexample with a smooth strictly starlike boundary, using quantitative control of the two strict Cassinian inequalities.

Task 3. Determine the sharp universal small-radius guarantee. Determine the largest universal constant a, if one exists, such that $r \operatorname{dist}(x, \partial G) \leq a$ guarantees starlikeness of $B_{c_G}(x, r)$ for every starlike domain G centered at x.

Task 4. Characterize centers with radial monotonicity. Characterize geometric conditions on (G, x) that ensure $t \mapsto c_G(x, x + tv)$ is nondecreasing along every segment in G.

Task 5. Classify the cone family. For $G = B(0, R) \setminus \{(u, v) : u \geq 1,\ |v| \leq b(u-1)\}$, determine the parameter and radius ranges producing nonstarlike Cassinian balls about 0.

References

[1] R. Klén, M. R. Mohapatra and S. K. Sahoo, *Geometric properties of the Cassinian metric*, arXiv:1511.01298v1 (2015), Definition on page 2 and Conjecture 4.7 on page 8. Published in Math. Nachr. 290 (2017), 1531–1543, doi:10.1002/mana.201600117.

[2] R. Klén, *Hyperbolic type distances in starlike domains*, arXiv:1603.00590v1 (2016), page 3.